

# Akilan K

Software Developer — Python, Cloud, AI/ML

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## Career Objective

Aspiring **Software Engineer** skilled in **Python, AI/ML, Cloud Computing, DevOps, and DSA**, with experience developing real-world projects and automation solutions. Seeking opportunities to build scalable software, solve complex problems, and grow as a technology professional.

## Education

**Madras Institute of Technology, Anna University** B.E. Computer Science Engineering (2022 – 2025)

**Nandha Polytechnic College** — Diploma in Petrochemical Engineering (2020 - 2022)

## Technical Skills

|                                      |                                                                    |
|--------------------------------------|--------------------------------------------------------------------|
| <b>Programming :</b>                 | Python, C/C++, SQL, OOP, DSA, Problem Solving                      |
| <b>Cloud &amp; DevOps:</b>           | AWS (Lambda, S3), Azure, Docker, Kubernetes, GitHub Actions, Linux |
| <b>AI/ML Frameworks &amp; Tools:</b> | TensorFlow, PyTorch, Scikit-learn, CNN, NLP, Transformers, EDA     |
| <b>Information Security:</b>         | Network Security, Blockchain Security Concepts                     |

## Projects and Experience

- **Automatic Model Selection for ML (2025):** Built a multi-model machine learning framework using **Python, Scikit-learn, Pandas, and custom NLP** that analyzed data characteristics, selected optimal algorithms, and automated model optimization; reduced manual model experimentation while improving efficiency and decision-making in predictive analytics. [GitHub](#)
- **Blood Cell Classification (2024):** Limitations in manual blood cell analysis by developing a CNN-based image classification model using **Python, TensorFlow, CNN, and Flask** implemented preprocessing and feature extraction pipelines to improve image quality and model performance, achieving **88–92% classification accuracy** with real-time reporting capabilities. [GitHub](#)
- **Community Detection Framework (2024):** Engineered a graph-based analytics framework using **Python, Graph ML, Network Analysis, and Data Visualization** to examine complex network structures discovered community clusters and key influencers, enabling deeper understanding of user interactions and network behavior. [GitHub](#)
- **FORTICRYPT – Five-Layer Hybrid Security System (2024):** Engineered a five-layer hybrid security framework using **HTML, JavaScript, Cybersecurity, and Encryption** to mitigate web and cloud security vulnerabilities implemented encryption, monitoring, and threat detection capabilities, strengthening protection against potential attacks and unauthorized system access. [GitHub](#)

## Achievements and Certifications

- Reported a security vulnerability involving **session hijacking and cookie theft** through HackerOne.
- Earned the **GitHub Quickdraw Achievement**, recognizing fast collaboration in GitHub workflows. [\[View\]](#)
- Contributed to **open-source projects** in ML, Data Engineering, and Cybersecurity via GitHub. [\[View\]](#)
- Earned **Certificate** in **EY Techathon 5.0 (Round 1 – Executive Summary Submission)**, demonstrating problem-solving, business analysis, and technology-driven innovation skills.

## Profiles

Actively practicing on coding and security platforms:

- [LeetCode](#) — Problem solving (DSA practice)
- [HackerRank](#) — Coding and algorithm practice
- [HackerOne](#) — Web security vulnerability research